

according to thoughts. Thus, a neural impulse is generated in the motor cortex of the brain. This movement of generation of a neuron signal is sensed by a microelectrode, which can filter out unwanted signals or noise.

This signal is carried by a connector to a signal amplifier, which amplifies it accordingly and sends to a neural signal interpreter using a fibre-optic cable. The neuron signal interpreter converts this signal or intention of the user into an electrical signal or computer commands. The computer then works according to the commands and translates these into actions (Fig. 2). The brain gate system can sense many thoughts at a time. It is capable of recording and amplifying numerous intentions of the user in the form of electrical signals.

Methods for sensing neuron signals

Basically, there two methods for sensing neuron signals: electrocorticography (ECoG) and electroencephalography (EEG).

ECoG. This is an invasive method for sensing a neuron signal. In this method, during surgery a chip is placed into the brain. The chip measures the neuron signal flow.

This method requires invasive surgery. However, the most challenging task is that, the chip in the brain, if left for a long time, can damage nearby tissues. This can even cause infections.

It has major advantages, too. Some of these are high spatial resolution, high frequency range, better signal-to-noise ratio and less complexity.

EEG. This non-invasive method (Fig. 3) for sensing a neuron signal does not require surgery. A set of electrodes are attached to the scalp. Since the electrodes are not directly attached to the motor cortex of the brain, some signals are blocked by the skull. This can cause distortion. In this method, temporal resolution

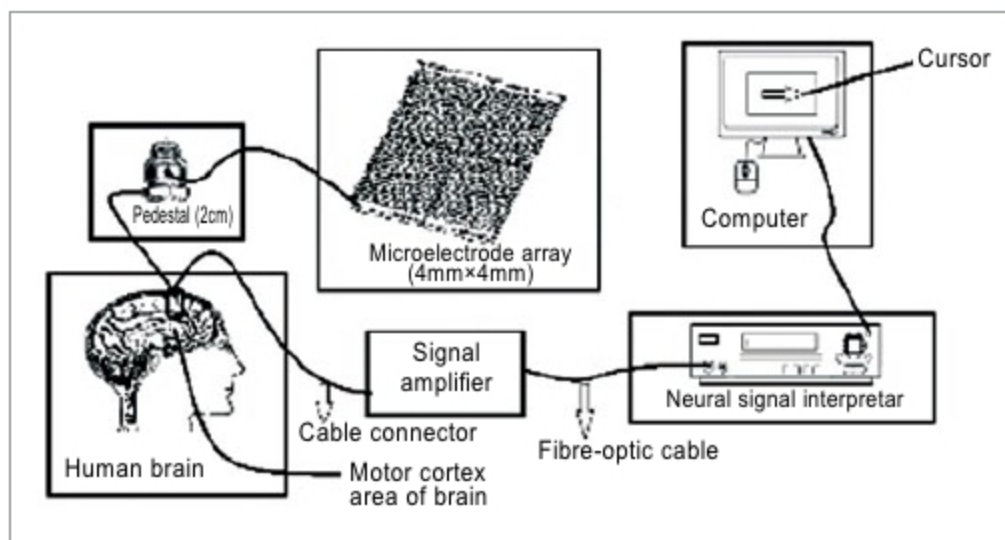


Fig. 2: Brain gate system

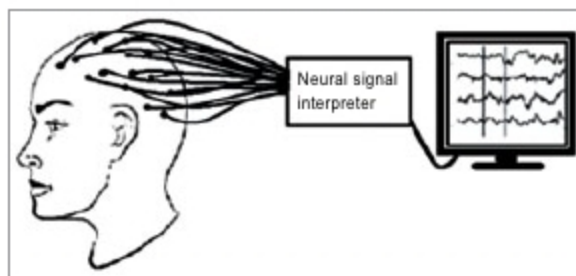


Fig. 3: EEG for sensing neuron signals

is fine, is easy to design, has good portability and low set-up cost. But, it has poor signal resolution due to the dampening of signals by the skull. EEG implantation is carried out by a specialist.

Achievement so far

After its development in 2003, several patients have been implanted with brain gate technology. According to the data proposed by Cyberkinetics and Department of Neuroscience, Brown University, most cases have been of spinal cord injury, while only a few cases were related to advance ALS.

Brain gate project has been approved by Investigational Device Exemption (IDE) of the US. This means that this project is valid for pre-market clinical trials.

Scope and applications

Brain gate technology can sense, record, transmit, analyse and amplify simultaneous movements of a neuron signal. In this technology, a brain gate neural interface system works faster than what is possible with the hands of a person. Using brain gate technology, multitasking is also possible. This means that this device can be used while performing other tasks simultaneously.

Accuracy is not affected by surrounding environment activity. The technology can also be used in a variable environment.

Brain gate allows people to move robotic arms with the help of electrical signals produced by the neural interfaced unit. Using this technology, physically-disabled people can operate a computer

through their thoughts and perform computer-based tasks like reading e-mails and moving the cursor.

Brain gate improves the quality of life for physically-disabled people, by allowing them to control TV channels, wheelchairs and so on, through their thoughts. It allows them to work in any industry/organisation.

Challenges

Brain gate technology has the following challenges:

1. It is difficult to synchronise the speed of thinking of a person and cursor/robotic arm movement.
2. Experiments or trials can be performed with the help of this technology as it is approved by IDE. But, full application on humans is not approved by Food and Drug Administration (FDA), USA.
3. Surgery is quite risky and costly.
4. Information transfer rate is limited.
5. To make the system wireless that will help people be more productive and independent needs work.

Future expectations

Cyberkinetics and Department of Neuroscience, Brown University hope to hone brain gate in upcoming years to make its pedestal more compact, and to make it completely wireless. Brain gate project is hopeful to get approvals from Department of Health and Human Services and FDA in the coming years.

Brain gate has good potential towards brain mapping, storage of memory and advanced humanoid robots. **EFY**